

Unit

3

Lesson

3



Beyond Imagination.

Circuits in Control!

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Observe and test a simple LED circuit to see if it works correctly.
- Troubleshoot the circuit by checking the power, wire connections, LED direction, and resistor placement.
- Learn that the long leg of the LED is the positive side and must connect to the correct spot.
- Understand that the resistor helps control the flow of electricity so the LED doesn't burn out.

Challenge questions of the lesson

1. How can you tell if an LED circuit is working correctly? What should you look for?
2. If the LED doesn't light up, what are some things you can check to find the problem?
3. Why is it important to connect the long leg of the LED to the positive side? What happens if you don't?
4. What role does the resistor play in the circuit, and why can't we just connect the LED directly to the battery?

SEL Relation

- Self-Awareness: Recognize and express feelings like curiosity and frustration during circuit building.
- Self-Management: Practice patience and persistence while troubleshooting and testing circuits.
- Social Awareness: Develop empathy by understanding mistakes are part of learning.
- Relationship Skills: Communicate and collaborate respectfully with peers during activities.
- Responsible Decision-Making: Make safe, thoughtful choices when fixing and testing circuits.

Engage



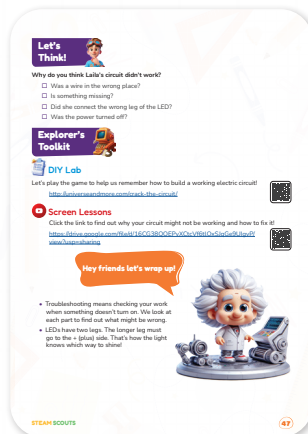
Teacher's Role:

- **Story Facilitator:** Begin the lesson by reading aloud the story of Lila's attempt to connect a light circuit. Use expressive voice tones to emphasize Lila's confidence, her troubleshooting moment, and Ms. Sara's encouraging response.
- **Inquiry Provoker:** Pause after the story and ask guiding questions to spark curiosity, such as "Why do you think the LED didn't light up?" or "Have you ever tried something and had to fix it?"
- **SEL Supporter:** Highlight the growth mindset theme by acknowledging effort, mistakes, and persistence. Reinforce that it's okay when things don't work the first time.

Expected from Students:

- **Active Listening:** Follow along with the story and pay attention to the characters' actions and feelings.
- **Reflection and Prediction:** Participate in the "Think" box activity by selecting possible reasons Lila's circuit might have failed.
- **Discussion Participation:** Share ideas with the class about troubleshooting circuits and relate them to their own experiences, if any.

Parts of the books included.



Story



1. The teacher gathers the students and says:

- "Let's see what Lila is up to today! She remembers how to build something using electricity—but something didn't go as planned. Let's read and figure out what happened, and how we can help her solve it!"

2. The teacher reads the story aloud with emotion—emphasizing Lila’s confidence, her challenge, and Ms. Sara’s supportive response. This models a growth mindset and encourages persistence.

Possible Strategies for Implementation:

- Interactive Read-Aloud with Think Alouds: Pause during the story to ask students reflective questions like:
 - “What do you think will happen when Lila presses the button?”
 - “Have you ever tried something that didn’t work right away? What did you do?”
- Empathy Building: Ask students how Lila might be feeling and why. Connect the situation to their own experiences with solving problems.
- Visual Demonstration or Roleplay (Optional): Show real circuit parts (wires, LED, battery) and let students guess what might go wrong—or invite two students to act out Lila and Ms. Sara’s conversation.
- Connect to Prior Knowledge: Ask students, “What do you remember about how circuits work?” and let them share ideas before continuing.
- Set the Purpose: Tell students, “Today, we’ll be engineers like Lila. We’ll learn how to fix and improve things using our coding and building skills!”

Let’s Think!



Objective:

Students will apply their understanding of basic circuit components and troubleshooting strategies to identify possible reasons a simple LED circuit might not work. This encourages analytical thinking and reinforces concepts related to power, connections, and polarity.

Answers:

- Was a wire in the wrong place?
- Is something missing?
- Did she connect the wrong leg of the LED?
- Was the power turned off?

All four options are technically correct and help students think critically about common circuit-building mistakes.

Possible Strategies for Implementation:

1. Guided Discussion:

- After showing the story and posing the Think question, prompt students to justify their answers:
 - “Why do you think this might be the reason?”
 - “What would happen if the LED is connected backward?”

2. Circuit Model Demonstration:

- If possible, use a physical or digital circuit model to show each mistake in action—e.g., intentionally flip the LED or remove a wire to replicate Lila's issue.

3. Think-Pair-Share:

- Have students discuss with a partner before sharing with the class. This builds confidence and allows peer learning.

4. Anchor Chart Creation:

- Together as a class, make a poster listing “What to check when your circuit doesn't work” and display it for future reference.

5. SEL Tie-In:

- Reinforce that making mistakes is part of learning, just like Lila. Use prompts like:
 - “How would you feel if this happened to you? What could you try next?”

Explorer's Toolkit



Teacher's Role:

- Facilitator of Prior Knowledge:
- Introduce the digital game as a way to review what students already know about electric circuits. Say something like:
- “Let's see if we can crack the circuit and spot what's wrong just like Lila!”
- Guide and Observer:
- Monitor students while they play the game, checking for understanding. Pause to ask probing questions like:
 1. “What happens if you leave out the wire?”
 2. “Why did that circuit not light up?”
- Encourager:
Acknowledge effort and progress. Emphasize that trying different combinations helps deepen their understanding of circuits.

Expected From Students:

- Interactive Review Participation:
- Students will play the online game Crack the Circuit to identify errors in electric circuits.
- Observation and Reflection:
- Students will use trial and error to troubleshoot circuit errors and build a functional circuit within the game.

- Collaborative Thinking:

Students may work in pairs or small groups, discussing their strategies and helping one another understand why certain circuits work and others don't.

Parts of the books included



DIY Lab

Let's play the game to help us remember how to build a working electric circuit!

<http://universeandmore.com/crack-the-circuit/>



DIY Lab

Objective:

Students will review and reinforce their understanding of electric circuit components and how to troubleshoot common circuit issues by interacting with a digital simulation game.

Materials Needed:

- Computers with internet access
- Headphones (optional, if audio is included in the game)

Activity link:

<http://universeandmore.com/crack-the-circuit/>



Activity Instructions:

1. Direct students to open the game using the link provided.
2. Ask them to try solving as many circuit puzzles as they can in a set time (e.g., 15–10 minutes).
3. Encourage them to notice why certain circuits don't work and how they fixed them.
4. After playing, lead a class discussion:
 - “What were some common mistakes in the broken circuits?”
 - “What did you learn that can help Lila in the story?”

Possible Strategies for Implementation:

- **Think-Aloud Modeling:** Before students begin, project the game and model solving the first puzzle aloud so students understand how to approach it.
- **Partner Play:** Pair students to encourage collaboration, allowing them to explain reasoning to each other while solving.
- **Challenge Mode:** Offer a fun incentive—who can solve the most circuits in 10 minutes?
- **Connection to Real Circuits:** After the activity, compare virtual circuits to real-life LED circuit setups students will build later. Ask, “How is this like Lila's smartpad activity?”



Teacher's Role:

- **Facilitator:** Guide students through the video with prompts before, during, and after watching.
- **Clarifier:** Address misconceptions about how circuits work and help students identify common troubleshooting techniques.
- **Connector:** Help students relate the video content back to the earlier story about Lila and her non-working LED.

Expected From Students:

- Watch the video attentively.
- Identify at least two reasons why a circuit might fail.
- Participate in a short discussion about what they learned.
- Apply this knowledge to fix or build their own circuits later in the lesson.

Parts of the books included.



Screen Lessons

Click the link to find out why your circuit might not be working and how to fix it!

<https://drive.google.com/file/d/16CG38QOEPvXCtcVf6tlOxSJqGe9UlqvP/view?usp=sharing>



Screen Lessons

Objective:

Students will understand common reasons why an electric circuit may not function and explore basic troubleshooting methods to identify and fix the problem.

Video link:

<https://drive.google.com/file/d/16CG38QOEPvXCtcVf6tlOxSJqGe9UlqvP/view?usp=sharing>



Possible Strategies for implementation:

1. Pre-Watching: Set a Purpose

- **Strategy:** Prompt-based Prediction
- **How to Implement:**
 - Before playing the video, ask students:
“What do you think could go wrong when building a circuit like Lila’s?”
 - Write down a few answers on the board so students know what to look for while watching.

2. During Watching: Active Engagement

- **Strategy:** Visual Checklist
 - Give students a simple checklist with icons (e.g., wire, LED, power) and ask them to tick off the issues mentioned in the video.
- **How to Implement:**
 - Pause at key moments to discuss each potential error and clarify any confusion.

3. Post-Watching: Discussion and Reflection

- **Strategy:** Think-Pair-Share
- **How to Implement:**
 - Ask students to turn to a partner and answer:
“If Lila asked you for help, what’s one thing you would check in her circuit?”
 - Then share some answers with the whole class.

Elaborate and Evaluate



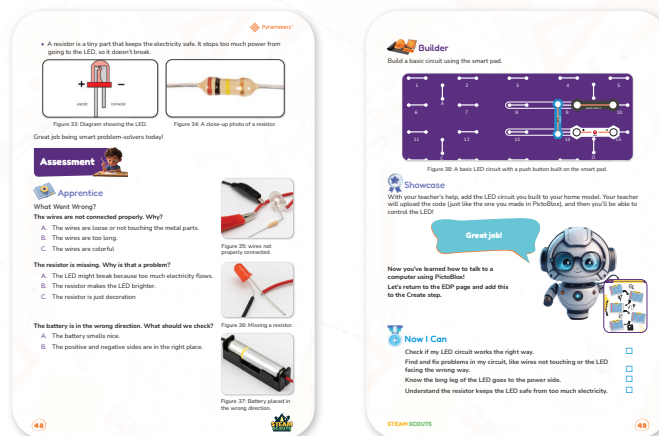
Teacher’s Role

- **Observer and Feedback Giver:** The teacher monitors students as they build their circuits using the smart pad. Watch how students connect components and troubleshoot any issues.
- **Guide Inquiry:** Encourage students to use critical thinking when identifying what might have gone wrong in faulty circuits (as shown in the first image). Prompt them with questions instead of giving direct answers.
- **Promote Collaboration:** Pair or group students to discuss their designs and identify how to improve their circuits together.

Expected From Students

- **Hands-On Circuit Assembly:** Students will build a working circuit using the smart pad and components such as wires, LEDs, resistors, and batteries.
- **Error Identification:** Using the “What Went Wrong?” guide, they must identify and explain possible circuit errors (e.g., incorrect wire placement, missing resistors, reversed batteries).
- **Problem Solving and Correction:** Students will work to fix any issues in their setup, test their circuits again, and explain what changes they made to get it working.

Parts of the books included.



Apprentice

Objective:

Students will analyze common circuit mistakes to understand how to troubleshoot and correct them. They will apply their knowledge of circuit components (wires, resistors, batteries) to identify causes of failure and suggest accurate solutions.

Answers:

1. The wires are not connected properly. Why?

A. The wires are loose or not touching the metal parts.

2. The resistor is missing. Why is that a problem?

A. The LED might break because too much electricity flows.

3. The battery is in the wrong direction. What should we check?

B. The positive and negative sides are in the right place.

Possible Strategies for Implementation:

• Pair and Share Discussion:

- Students can work in pairs to analyze each image and discuss what might be wrong before selecting an answer. This encourages collaboration and verbal reasoning.

• Circuit Detective Roleplay:

- Assign students the role of "Circuit Detectives" where they inspect the image "cases" and document what went wrong and why. They can write their conclusions in a notebook or on a mini report form.

• Real-Time Demonstration:

- Use real circuit components in class to recreate the problems shown. Let students fix each mistake themselves after analyzing the diagram.

• Connect to Previous Learning:

- Before the activity, review the roles of resistors, polarity of batteries, and importance of metal-to-metal contact using a brief demo or animated clip.

• Exit Ticket:

- At the end, ask students to reflect: “What was the most important thing you learned about fixing circuits today?”



Builder

Objective:

Students will demonstrate understanding of basic circuit concepts by constructing a simple, functioning circuit using a smart pad. They will apply knowledge of polarity, connectivity, and components (battery, LED, wires) to ensure the circuit works properly.

Materials Needed

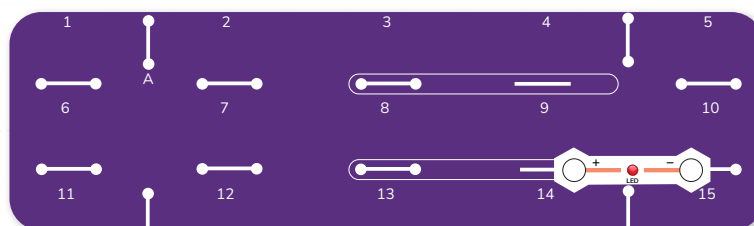
- Smart Pad
- LED light
- Resistor (for component safety)

Activity Instructions

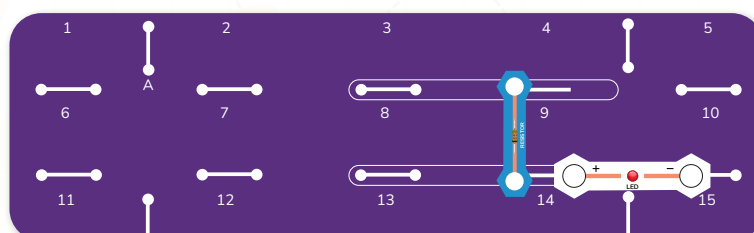
1. Provide each student or group with a Smart Pad and all required circuit components.
2. Ask students to plan how they will place the components to create a complete path for electricity.
3. Guide them to insert the button, resistor, and LED into the Smart Pad.
4. Prompt them to test the circuit—check if the LED lights up.
5. If it doesn't work, encourage students to troubleshoot by reviewing the orientation of the LED, battery placement, and wire connections.
6. Once successful, students can sketch their final working circuit and label each component.

Correct step by step guide:

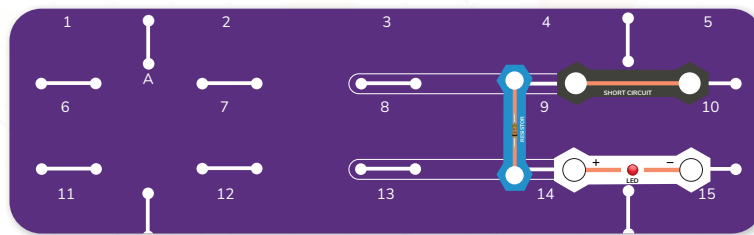
1. Connect the battery's positive end with the LED's positive end (the taller leg).



2. Connect the LED's negative end (the shorter leg) to a resistor's terminal.



3. Connect the other terminal of the resistor to the battery's negative end.
4. Connect smart pad with your power source using power source and see what happens.



Possible Strategies for Implementation

1. Model First, Then Release

- **How to implement:** Begin by demonstrating how to place one component (e.g., the resistor) on the smart pad without fully solving the circuit. Then allow students to try the rest themselves. This provides guidance without giving away the solution.

2. Use Peer Pairing

- **How to implement:** Pair students strategically (e.g., one student confident with circuits and one still learning). Let them problem-solve together, encouraging communication and shared decision-making.

3. Circuit Debugging Cards

- **How to implement:** Prepare small cards with common circuit issues (e.g., “Check LED direction,” “Are wires loose?”). If a group’s circuit doesn’t work, let them draw a debugging card and investigate. This builds diagnostic thinking.

4. Use “I Notice, I Wonder”

- **How to implement:** Before testing, ask students to write or share one thing they notice about their circuit and one thing they wonder (e.g., “I notice the LED is not lighting. I wonder if the battery is placed correctly.”). This promotes metacognitive thinking.

5. Encourage Safe Risk-Taking

- **How to implement:** Assure students that it's okay if their circuit doesn't work the first time. Praise effort and persistence. Remind them that real engineers troubleshoot and revise often.

6. Anchor to Real-World Use

- **How to implement:** Ask students how their circuit could help someone in real life (e.g., powering a nightlight for someone who can't use a switch). This builds empathy and connects science to purpose.

7. Support With Visual Aids

- **How to implement:** Post a labeled diagram of a sample circuit on the board as a reference—not a copy—but a scaffold for visual learners.

Elaborate and Evaluate



Teacher's Role:

- **Circuit Integrator:** Safely connect and upload the LED code to the Arduino on behalf of students.
- **Facilitator:** Guide students through how their circuit will be controlled through code (e.g., made in PictoBlox).
- **Discussion Leader:** Encourage students to share what they learned about how circuits work and how they can be used in smart homes.

Expected from Students:

- Insert or attach the LED circuit into their cardboard/model home in a clear location.
- Observe what happens when the code is uploaded.
- Explain what the LED circuit does, how it was built, and how it fits into their smart home.
- Reflect on what they would improve or change.



Showcase

Objective:

Students will demonstrate their understanding of circuit building by integrating their LED circuit into their home model and observing how it functions when programmed. They will explain how the circuit works and its real-life application.

Materials Needed

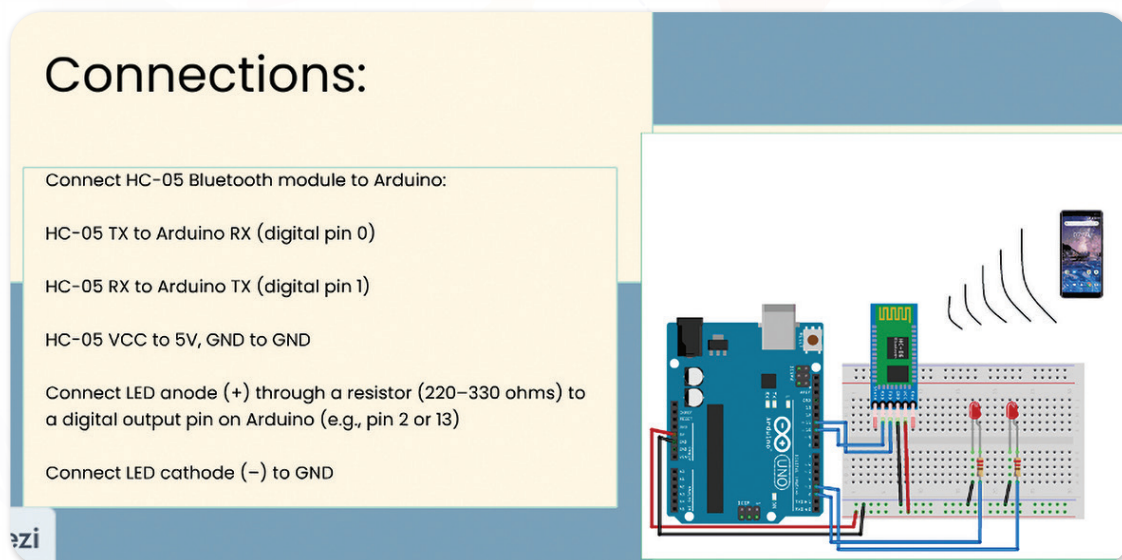
- Arduino Board+ Cable (e.g., Arduino Uno)
- Bluetooth Module (e.g., HC-05 or HC-06)
- LEDs (for demonstration)
- Resistors (220 ohm or 330 ohm)
- Breadboard
- Connecting Wires
- Smartphone with Bluetooth

Resources: https://drive.google.com/drive/folders/1FdUxih_I1TP5o_WkdFW_160DuTEmTdK8?usp=drive_link



Activity Instructions:

1. Prepare the Arduino setup ahead of time with preloaded test code (blinking LED).



2. As students finish their home models and insert the LED, connect each group's LED leads to the Arduino board (teacher only).
3. Upload the code while students watch.

```
#include <SoftwareSerial.h>
```

```
SoftwareSerial BLU(0, 1); // RX, TX
```

```
String voice;
```

```
int Red = 2; // Connect to Pin #2
```

```
int Green = 3; // Connect to Pin #3
```

```
int Blue = 4; // Connect to Pin #4
```

```
void setup() {
```

```
  Serial.begin(9600);
```

```
  BLU.begin(9600);
```

```
  pinMode(Red, OUTPUT);
```

```
  pinMode(Green, OUTPUT);
```

```
  pinMode(Blue, OUTPUT);
```

```
}
```

```
void loop() {
```

```
  if (BLU.available()) {
```

```
    voice = BLU.readString();
```

```
if (voice == "turn on red") {  
  digitalWrite(Red, HIGH);  
} else if (voice == "turn off red") {  
  digitalWrite(Red, LOW);  
} else if (voice == "turn on green") {  
  digitalWrite(Green, HIGH);  
} else if (voice == "turn off green") {  
  digitalWrite(Green, LOW);  
} else if (voice == "turn on blue") {  
  digitalWrite(Blue, HIGH);  
} else if (voice == "turn off blue") {  
  digitalWrite(Blue, LOW);  
}  
}  
}
```

4. Ask each group to describe:

- What the LED does
- Where they placed it in the house and why
- What they learned about electricity and design

Possible Strategies for Implementation:

1. Gallery Walk

- Let students walk around and view each other's houses and LED placements.
- Provide sticky notes for compliments or questions.

2. Reflective Sharing Circle

- Each group presents briefly to the class about their process, successes, and one challenge they solved.

3. Coding Connection

- Show students the code on the board and explain briefly how "turn on" and "turn off" blocks work so they can connect the physical to the digital logic.

4. Visual Cues and Labels

- Provide stickers or cards for students to label the purpose of the LED in their house (e.g., "bedroom light," "security light").

5. Photo Documentation

- Take a picture of each model with the LED working and display on the classroom wall or digital board as part of a "Smart Home Showcase."

Rubric – My Smart LED Home!

What We Look For	Needs Help (1–5 pts)	Getting There (6–10 pts)	Great Job! (11–15 pts)	Wow! You Nailed It! (16–20 pts)
Where is the Light?	I didn't add the LED or it doesn't fit my house.	I added the LED but it's in a random spot.	I placed the LED where it makes sense, like in a room.	My LED is in a smart spot (like the porch or bathroom), and it really shows my idea!
What Does It Do?	I didn't explain what the light is for.	I said what it is, but not clearly.	I told what the light does in my house.	I explained clearly and gave a smart reason for using the light!
How Does It Work?	I didn't say how it works.	I tried, but I was a little confused.	I said it needs power and showed how it turns on.	I explained clearly, like a real engineer—talked about wires, power, or the smartpad!
Team and Talk	I didn't help or talk about my project.	I helped a little but needed reminders.	I shared my part and helped my team.	I spoke clearly, helped others, and worked like a superstar teammate!
How My House Looks	I rushed and didn't try my best.	I added a few things, but not much.	My house looks neat and shows my idea.	My house is creative and shows a smart idea with great effort!

It's time to move to the create part and add this.

Create

We added a small LED light to our smart home model. I decided to put the light in the kitchen because that's where people need to see clearly when cooking. I didn't build the circuit myself, but my teacher helped connect it to the smartpad.

I explained to my friends how the LED works when the smartpad sends electricity to it. It was fun to see it light up! I wanted my house to be safe and bright, just like a real smart house.



Now I Can

Objective:

Students will reflect on their learning by identifying what they know about building and checking simple LED circuits. They will demonstrate understanding of basic troubleshooting, LED polarity, and resistor safety in an electric circuit.

Activity Instructions:

- **Read Together:** Read each “Now I Can” statement aloud with the class. Pause to explain each idea using real examples from your build time.
- **Self-Check:** Give students a printed copy or display the checklist on the board. Ask students to put a check next to each skill they feel confident about.
- **Peer Talk:** Let students pair up and explain one “Now I Can” item to a partner, using their own project as an example.
- **Wrap-Up Discussion:** Ask a few students to share which part was the trickiest and how they fixed it.

Relation to Standard

1. NGSS (Next Generation Science Standards):

PS3.B: Conservation of Energy and Energy Transfer

Students explore how electrical energy can be transferred to produce light (via LED) and how circuit components like resistors affect this transfer.

ETS1.A: Defining Engineering Problems

When the circuit doesn't work, students identify and analyze the problem (e.g., incorrect wiring, LED polarity), then brainstorm and test solutions.

ETS1.B: Developing Possible Solutions

Through trial and error, students learn how to improve their designs and explain why a solution worked, aligning with the engineering design process.

2. ISTE Standards for Students:

Empowered Learner (1c):

Students use digital tools (e.g., PictoBlox simulation or teacher-led uploads) to troubleshoot and understand circuit behavior.

Knowledge Constructor (3d):

Students build knowledge by experimenting, fixing, and reflecting on real-world electrical components.

Innovative Designer (4a–4d):

Students apply a design thinking approach: they plan, build, test, and revise their circuit designs to solve a practical problem.

Unit

3

Lesson

4



Beyond Imagination. Sound Explorers.

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Identify that sound is made by vibrations.
- Describe how we can feel sound through movement.
- Observe sound moving through classroom objects.
- Use voice commands to turn on/off an LED.

Challenge questions of the lesson

1. What makes sound, and how do we know it's moving?
2. Can we feel sound, even if we can't see it?
3. How does sound travel through things in our classroom?
4. Can we use our voice to control things like lights?

SEL Relation

- **Self-Awareness:** Express excitement and pride in learning how sound helps others.
- **Self-Management:** Practice focus and patience during experiments and activities.
- **Social Awareness:** Show empathy by understanding how technology supports people's needs.
- **Relationship Skills:** Communicate, listen, and collaborate with classmates.
- **Responsible Decision-Making:** Reflect on using sound and technology responsibly to help others.

Engage



Teacher's Role

- **Story Facilitator:** Read the scene aloud or play an audio version. Use tone, gestures, and pauses to bring the story of Lila and Uncle Sam to life. Emphasize emotional shifts (e.g., Uncle Sam's excitement or Lila's pride) to help students connect.
- **Discussion Leader:** Pause after key moments to ask comprehension and reflection questions. Encourage students to predict what will happen next or make connections to real-life uses of voice technology.
- **Think Question Guide:** Lead students through the "Let's Think!" question. Read each choice and let them vote or discuss in pairs before sharing their reasoning. Prompt deeper thinking by asking "Why do you think that's the right answer?"

Expected From Students

- **Active Listening:** Pay attention to the story and respond to questions about how Uncle Sam feels and what the voice commands do.
- **Comprehension & Reflection:** Complete the "Story Reflection" activity by choosing correct answers that show understanding of the narrative.
- **Critical Thinking:** Analyze how voice and sound are used to control smart objects using the "Let's Think!" question. They should explain their answers in their own words.

Parts of the books included



Narrator: The big day had come! Uncle Sam was ready to try out his new smart home. Lila and her family worked hard to set up all the voice controls she had planned.

Uncle Sam: "Wow, Lila! This is amazing! Lights on!"
(The lights turned on, and Uncle Sam smiled happily.)

Lila: "Now you can use your voice to control everything, Uncle Sam!"

Uncle Sam: "This is so cool! I can do things by myself again."

(He moved around the room. He used his voice to open the curtains, turn on the radio, and change the lights.)

Narrator: Uncle Sam felt happy and proud. The smart home-made things easier and helped him feel closer to his family.

Uncle Sam: "Thank you, Lila! My home is better now, and I can do more on my own."
(Lila smiled, feeling proud that she helped her uncle.)

Story Reflection

1. How did Uncle Sam feel when he first used the smart home?
 - A. Happy and excited.
 - B. Sad and tired.
 - C. Angry and confused.

STEAM SCOUTS

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Pyramakers™

2. What are some things Uncle Sam could control by talking?
 - A. The lamp.
 - B. The garden and the car.
 - C. The fridge and the washing machine.
3. Why did Lila work hard to make Uncle Sam's home smart?
 - A. To help Uncle Sam live more easily.
 - B. To make the house bigger.
 - C. To throw a party.

Let's Think!



How does your voice make sound, and how can sound be used in a smart home?

- A. Your nose makes sound, and sound changes light colors.
- B. Your voice comes from shaking parts in your throat, and sound can turn things on.
- C. Your teeth make sound, and sound opens windows.
- D. Your eyes blink, and smart homes use buttons only.

Explorer's Toolkit



DIY Lab

Let's See Sound in Action!

- Cover the top of the bowl with plastic wrap.
- Use a rubber band to keep the plastic wrap tight.
- Sprinkle a few sprinkles or tiny paper bits on top.
- Bring your lips close to the side of the bowl (but don't touch it).
- Hum loudly and watch the sprinkles carefully!
- Try humming louder or changing your pitch (high or low) if nothing moves.



Figure 39: A DIY sound experiment showing a bowl covered with plastic wrap and sprinkles that move when sound vibrations are made nearby.

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STEAM SCOUTS



1. Set the Context:

- Begin by reminding students of Lila and Uncle Sam from previous lessons. Say something like:
 - “Remember how Lila was trying to help Uncle Sam with a smart home? Today is the big day—let’s find out what happens when he tries it for the first time!”

2. Create Curiosity:

- Ask a question to activate thinking:
 - “What do you think it would feel like to control your home using just your voice?”
- Invite students to share quick guesses or experiences with voice assistants (e.g., Alexa, Siri).

3. Read the Story Expressively:

- Read the scene with expression and dramatic pauses. Use gestures to act out moments (e.g., turning on lights, speaking commands).

Possible Strategies for Implementation:

1. Role-Play Reading:

- Assign students to play each character (Narrator, Lila, Uncle Sam) and perform the scene in front of the class. This boosts comprehension, fluency, and engagement.

2. Story Mapping:

- After reading, work together to fill out a simple “Feelings Map”:
 - What problem did Uncle Sam have?
 - What solution did Lila create?
 - How did Uncle Sam feel before and after?

3. Prediction Pause:

- Stop midway (before Uncle Sam uses voice commands) and ask students to predict what will happen next.
- Example prompt: “What do you think he’ll say? What might the house do?”

4. Think-Pair-Share:

- After reading, ask: “How did the smart home help Uncle Sam feel more independent?” Let students discuss with a partner, then share as a class.

5. Anchor to Real Life:

- Ask students if they’ve seen or used smart devices before. Invite them to connect the story to real-world examples.

Story Reflection



Objective:

Students will reflect on the story of Uncle Sam's smart home and answer questions that demonstrate their understanding of how voice-controlled technology can support people's independence and well-being. This activity helps students connect emotions, actions, and technology use, fostering empathy, comprehension, and real-world application of smart solutions that promote accessibility at home.

Answers:

Story Reflection



1. How did Uncle Sam feel when he first used the smart home?

☒ A. Happy and excited

☐ B. Sad and tired.

☐ C. Angry and confused.

2. What are some things Uncle Sam could control by talking?

☒ A. The lamp.

☐ B. The garden and the car.

☐ C. The fridge and the washing machine.

3. Why did Laila work hard to make Uncle Sam's home smart?

☒ A. To help Uncle Sam live more easily

☐ B. To make the house bigger

☐ C. To throw a party

Possible Strategies for Implementation:

1. Guided Reading Review

- **How to implement:** Re-read or revisit parts of the story together. Pause at key points where answers are revealed. Ask students to raise their hands or signal when they hear the answer.

2. Think-Pair-Share

- **How to implement:** For each question, let students first think quietly, then discuss their answer with a partner before sharing with the class. This encourages participation and clarifies understanding.

3. Use Visuals and Gestures

- **How to implement:** Reenact scenes with student volunteers or use gestures to recall moments (e.g., turning on the light, smiling with pride). Then connect them to the reflection questions.

4. Circle the Correct Answer

- **How to implement:** Provide students with printed copies and ask them to circle their answers independently. Review together with explanations for each answer.

5. Character Voice Role-Play

- **How to implement:** Ask students to pretend to be Lila or Uncle Sam and answer a question "in character." For example, "Uncle Sam, how did you feel when the smart home worked?" This deepens empathy and understanding.

Let's Think!



Objective:

Students will explain how sound is produced by vibrations and how voice can be used as an input to operate devices in a smart home. This activity helps students link basic science concepts of sound with real-world applications in accessible technology.

Answers:

Correct Answer:

- B. Your voice comes from shaking parts in your throat, and sound can turn things on.
- A. Incorrect – the nose is not the main organ responsible for sound production.
- C. Incorrect – teeth do not produce sound.
- D. Incorrect – smart homes often use voice, not blinking eyes, for control.

Possible Strategies for Implementation:

1. Think-Pair-Share:

- **How to Implement:** After showing the question, ask students to first think silently for 30 seconds. Then, pair up and share their ideas with a partner before answering. Follow with a class discussion to review answers and explain the science behind voice commands and vibrations.

2. Use a Hands-on Demonstration:

- **How to Implement:** Let students gently place their hands on their throats and hum to feel the vibration. Link this tactile experience to the correct answer and explain how microphones in smart homes detect voice vibrations.

3. Connect to Real Life:

- **How to Implement:** Ask students if they've ever seen someone use "Hey Google" or "Alexa." Let them share what they think is happening when a device responds to voice. Use their answers to bridge into the correct choice.

4. Science Wall Chart Extension:

- **How to Implement:** Add a visual poster or chart in the class showing the throat, vocal cords, and how sound travels as vibrations. Revisit the chart during or after answering the question to reinforce understanding.

Explorer's Toolkit



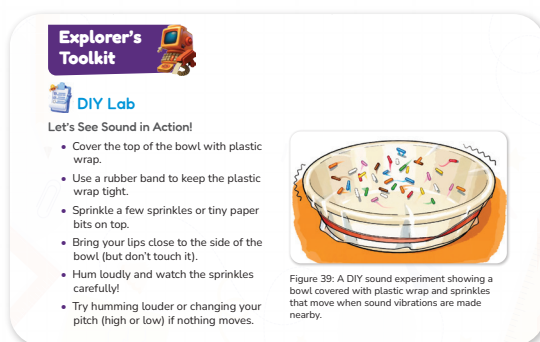
Teacher's Role:

- **Facilitator of Inquiry:** Set up stations and ensure all materials are ready. Prompt students to observe carefully and guide them without giving away the results.
- **Support Observer:** Encourage students to test different sounds and pitches, and ask probing questions like “What do you think caused that?” or “What changed when you hummed louder?”
- **Clarifier:** Help students understand the concept of vibrations by relating the movement of sprinkles to how sound travels through air.

Expected from Students:

- Follow instructions to set up the experiment.
- Make careful observations about the movement of the sprinkles.
- Experiment with different pitches and volumes.
- Record their observations and reasoning for what caused the movement.
- Discuss the results with peers and connect to the idea of sound vibrations.

Parts of the books included



DIY Lab

Objective:

Students will investigate how sound is caused by vibrations by observing how their humming creates movement in nearby objects. This hands-on experiment allows learners to explore invisible sound waves and connect them to real physical effects.

Materials Needed:

- A bowl
- Plastic wrap
- Rubber band
- Sprinkles or small paper bits
- Quiet classroom space for observation

Activity Instructions:

1. Cover the top of the bowl tightly with plastic wrap and secure it with a rubber band.
2. Sprinkle a few tiny objects (e.g., sprinkles or paper bits) on the plastic wrap.
3. Bring your mouth close to the bowl (without touching it).
4. Hum loudly and observe the movement of the sprinkles.
5. Try humming louder or changing your pitch if nothing moves.
6. Reflect by answering the follow-up questions:

1. What did you see happen when you changed your voice?

The sprinkles jumped a lot

2. What made the sprinkles move?

The sound from my humming

Possible Strategies for Implementation:

1. **Pair-Based Setup:** Students work in pairs—one hums while the other watches closely and notes the movement.
2. **Scientific Journaling:** Have students draw the setup in their notebook and record their predictions, results, and reflections.
3. **Use Visual Aids:** Show an animation or video of sound waves before or after the activity to deepen understanding.
4. **Mini Discussion Circles:** Pause mid-activity to let students share early observations and hypotheses.
5. **Connect to Real Life:** Ask students where they've seen or felt vibrations before (e.g., when sitting near a speaker or touching a running machine).



Teacher's Role:

- Introduce the concept that sound is made by vibrations, and prepare students to understand how this connects to real-world technology (like voice-controlled devices).
- Facilitate video viewing by setting a clear purpose, guiding key observations, and prompting reflection.
- Support vocabulary development (vibrations, sound waves, control, voice command).
- Clarify misconceptions and encourage students to share their ideas and questions.

Expected From Students:

- Watch the video actively and listen for how sound travels and how it can be used to control devices.
- Participate in pre- and post-video discussions.
- Use simple vocabulary to explain what they learned (e.g., "Sound moves," "It shakes things," "Voice can control lights").
- Begin connecting the science of sound to smart home applications.

Parts of the books included.

□ The bowl



Screen Lessons

Click on the link to watch the video and learn about sound!

<https://www.youtube.com/watch?v=m2mxVWrVP58>



Screen Lessons

Objective:

Students will explore how sound is made by vibrations and how these vibrations can be used in real-world applications such as smart homes. They will learn to observe and describe how voice commands use sound to trigger responses in electronic systems.

Video Link:

<https://www.youtube.com/watch?v=m2mxVWrVP58>



Possible Strategies for Implementation:

1. Pre-Watching: Set a Purpose

- **Strategy:** Activate prior knowledge and create curiosity.
- **How to Implement:**
 - Ask students, "Have you ever turned something on using just your voice?"
 - Write or draw student responses.
 - Tell students to watch the video to find out how sound helps smart homes work.

2. During Watching: Active Engagement

- **Strategy:** Prompt guided observation.
- **How to Implement:**
 - Pause the video at key moments to ask:
 1. "What is making the object move?"
 2. "Where is the sound coming from?"
 - Use gestures (like tapping the throat or miming vibrations) to help them visualize.



Apprentice

Objective:

Students will demonstrate their understanding that sound is produced by vibrations and can travel through different materials (like air and water). They will also observe and describe physical sensations that accompany sound production using their own voices.

Answers:

1. Sound is made when something

B. vibrates

1. Sound can travel through:

C. both A and B

Try It Out – I felt:

- A buzzing or vibrating feeling in their throat (free response; expected answer: “buzzing,” “moving,” “vibrating,” etc.)

Draw It!

Draw something that makes sound when it vibrates.



Possible Strategies for Implementation:

- **Model the Activity First:**
 - Begin by demonstrating the “Try It Out” section with the whole class. Say “Hello!” loudly and describe what you feel. Let students mirror you.
- **Think-Pair-Share:**
 - Have students think about what vibration means, pair up to test the activity, and share their answers.
- **Scaffold Vocabulary:**
 - Pre-teach or review key words like “vibration,” “travel,” and “buzzing” using visuals or real-life examples like guitar strings or loudspeakers.
- **Use Real-Life Examples:**
 - Connect the concept to everyday experiences—like feeling sound from a speaker or when talking while touching your throat.
- **Encourage Reflection:**
 - Ask open-ended questions like, “Why do you think you felt that?” or “What would happen if there were no air or water?”



Builder

Objective:

Students will explore how sound travels through solids by building a string phone. This hands-on activity helps students understand sound vibrations, communication methods, and the science of how sound moves from one place to another.

Additional resources link: <https://www.youtube.com/watch?v=PmmGSF7TXBI>



Materials Needed:

- 2 paper cups
- A long piece of string (1–2 meters)
- 2 paper clips or tape
- Something to poke holes (like a pencil)

Activity Instructions:

1. Poke a small hole at the bottom of each paper cup.
2. Thread one end of the string through the hole in the first cup and tie a paper clip or knot it to keep it in place.
3. Repeat on the other side with the second cup.
4. Two students should each hold a cup and walk apart until the string is tight.
5. One student talks into the cup while the other places their ear on their cup to listen.

Possible strategies to implement:

- **Pair and Practice:** Assign students in pairs and rotate so every student has a turn speaking and listening.
- **Guided Observation:** Ask guiding questions like:
 - “What do you feel if you touch the string?”
 - “Does the string have to be tight or loose for the sound to travel?”
- **UDL Tip:** Allow students to sketch or act out what’s happening inside the string. Use visuals to show sound waves traveling.
- **Scaffolded Support:** Give sentence starters: “I noticed that when I talked, the string...”, or provide labels for parts of the model (cup, string, sound waves).
- **Class Discussion:** Compare experiences and reinforce the idea that sound moves by vibration and needs a medium to travel.

Final Prototype:



Final prototype video:

<https://drive.google.com/drive/folders/1LZFl0M84ebjUlqqKPpyOk4fENppoSBbH>



Rubric – My Smart LED Home!

Criteria	Needs Improvement (1 pt)	Developing (2 pts)	Proficient (3 pts)	Excellent (4 pts)
Model Construction	Model is incomplete or not working	Cups are connected but string is too loose or tangled	String phone works with some help from teacher	String phone is fully functional and well built independently
Understanding of Sound Travel	Does not show understanding of how sound moves	Partial understanding shown through explanation or action	Understands that sound moves through string	Clearly explains that sound travels as vibrations through solids
Teamwork & Communication	Did not cooperate or share tasks	Sometimes cooperated but needed reminders	Worked well with partner, shared tasks fairly	Actively supported partner and worked together the whole time
Observation & Reflection	Gave no observations or ideas	Gave a basic answer with teacher prompting	Shared a clear observation of how the phone worked	Gave thoughtful observations and made connections to sound/vibration
Effort & Participation	Needed constant redirection	Participated with occasional reminders	Stayed focused and followed instructions	Fully engaged, asked questions, and showed curiosity throughout



Now I Can

Objective:

Students will demonstrate their understanding of how sound is made through vibration, how it can be felt and observed, and how voice commands can interact with technology like LEDs. This reflection helps reinforce key science and technology concepts introduced through hands-on activities.

Activity Instructions:

1. Display the “Now I Can” checklist on the board or distribute printed versions.
2. Read each statement aloud one at a time and ask students to put a checkmark if they feel confident about it.
3. Pause after each sentence and allow students to share an example or explain what they remember (e.g., “What did we use to see sound move?”).

4. Encourage peer sharing — students can explain to a partner how they completed a specific activity.
5. Celebrate learning progress and clarify any concepts that students are unsure about by revisiting related parts of the lesson.

Relation to Standard

1. Next Generation Science Standards (NGSS):

1-PS4-1:

Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate.

- Students observe how sound travels and makes objects move (e.g., sprinkles jumping).

3-PS2-2 (extension):

Make observations and/or measurements of an object's motion to provide evidence that a pattern can be used to predict future motion.

- The string phone shows how sound waves can travel predictably through a medium.

2. ISTE Standards for Students:

Empowered Learner (1a, 1c):

- Students set learning goals by checking “Now I Can” progress and using tools like voice-controlled LEDs.

Innovative Designer (4a, 4d):

- Students use design thinking to create a string phone and explore sound transmission.

3. CSTA Computer Science Standards (Grades K–2):

1A-CS-02:

Use appropriate terminology in identifying and describing the function of common physical computing components (e.g., LEDs, sensors).

- The LED activity reinforces understanding of inputs and outputs using voice recognition.

4. UN Sustainable Development Goals (SDG) Alignment:

SDG 4 – Quality Education:

- Promotes inclusive, hands-on, inquiry-based learning experiences.

SDG 9 – Industry, Innovation, and Infrastructure:

- Introduces smart tech and communication methods (voice commands, sound travel).

SDG 10 – Reduced Inequalities:

- Supports accessibility concepts (voice interfaces for assistive tech).



Mission Innovate

Design Another Smart Room

Overview:

This open-ended project challenges students to extend their understanding of inclusive smart design by creating a new tool for a different room in the home. Students will work as young engineers to make a space easier for someone who is blind, deaf, or has another disability. This encourages empathy, design thinking, and hands-on prototyping.

Project Duration: 1–2 sessions

Teacher's Role:

- Facilitator of Inquiry:
- Begin by reviewing the mission prompt with the class. Ask students to choose a room (not the same as in the first mission) and think about what's tricky for someone with a disability. Guide them to brainstorm problems and voice-activated or alternative solutions.
- Encourager of Empathy and Creativity:
- Lead a discussion using the critical thinking questions. Help students imagine how daily life is affected by blindness or hearing loss. Inspire them to think of kind, smart, and helpful features to support independence.
- Supporter of Student Work:
- Let students design and build independently or in pairs. Offer feedback only when needed, reminding them to apply what they've learned about voice control, recycled materials, and safety.
- Guide for Reflection and Communication:
- Before presenting, support students in explaining how their model helps people with disabilities. Use simple sentence starters if needed ("My design helps because..."). Invite respectful listening during presentations.
- Assessment Guide:
- Use a simple rubric (focused on design idea, usefulness, creativity, and clarity) to assess each team's final prototype.

Implementation Tips:

- Encourage students to build different rooms than the one they used in their first project.
- Offer recycled or classroom-safe materials like cardboard, straws, string, LED or buzzer kits, markers, etc.
- Help students brainstorm features like vibrating alerts, braille stickers, or sound-activated fans or switches.
- Support teamwork while allowing students to make individual design choices.
- Celebrate inclusive thinking and all ideas that focus on helping others.